Institutional Sources of Competitive Advantage and Selection into Competition

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Abstract

Competitive advantages may reflect ability or recognized qualifications, but also connections, inflated credentials, or unverifiable claims. We ask whether the institution producing these advantages changes who chooses to compete. Our randomized real effort experiment with 654 participants features equal task benchmarks and unequal positions created through computer assignment or private reporting. Relative to equal task benchmarks, an advantage assigned by computer raises entry by 13.0 percentage points, while a disadvantage lowers it by 15.3 points. Compared with computer assignment, private reporting raises the share who hold an advantage but choose piece rate by 12.2 percentage points, leaves the share who compete with an advantage unchanged, and reduces the share who compete from a disadvantage by 7.4 points. Among participants holding a competitive advantage, entry is 12.2 points higher after computer assignment than under private reporting, although selection may contribute to this difference. Limited verification therefore changes who competes and from which competitive position, even when total entry changes little.

Keywords: competition, private reporting, competitive advantage, gender difference.

JEL codes: C91, D91, J16, M52.

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1 Introduction

Competition determines access to jobs, promotions, grants, and procurement contracts, but contestants rarely begin from equal positions. Some advantages reflect demonstrated ability, recognized qualifications, or specialized training. Others arise through family connections, inflated credentials, or claims about past performance that are difficult to verify. The institution governing how an advantage is obtained can therefore shape who enters competition, even when the formal rules of the contest are unchanged.

Most research on competition entry treats the competitive position as given. A large literature explains entry through confidence, beliefs about relative performance, risk preferences, and preferences for competition, with particular attention to gender differences (see, e.g., Gneezy et al., 2003; Niederle and Vesterlund, 2007; Croson and Gneezy, 2009; Niederle and Vesterlund, 2011; van Veldhuizen, 2022). We instead study whether the way an advantage is obtained influences who competes. We compare positions created by an externally determined process with positions that follow private reports that may favor the participant.

We examine this question in a real effort experiment with 654 participants. Our design separates task difficulty, competitive advantage and disadvantage, and how those positions are determined. Participants choose between piece rate and tournament compensation. In Fixed Easy and Fixed Hard, matched participants perform the same task. In Random, the computer assigns easy and hard tasks by chance. In Die Roll, participants privately report a die outcome that determines their task, and individual reports cannot be verified. These four treatments separate the effect of competitive position from the effect of how it is determined.

We first establish that competitive advantage and disadvantage change entry in opposite directions. Relative to the corresponding equal task benchmark, an externally assigned advantage raises competition entry by 13.0 percentage points, a 22 percent increase relative to the Fixed Easy entry rate. An externally assigned disadvantage reduces entry by 15.3 percentage points, a 29 percent decrease relative to the Fixed Hard entry rate. Taken together, the opposing responses to advantage and disadvantage show why task difficulty and competitive advantage are not equivalent: what matters is not simply whether a task is easy or hard, but whether it improves or worsens expected performance relative to a competitor. These responses are driven mainly by men.

Our central result is that private reporting changes the joint distribution of competitive position and competition entry relative to random assignment. The share of participants who hold an advantage and choose piece rate is 12.2 percentage points higher, while the share who compete with an advantage changes by only 1.2 percentage points. At the same time, the share who compete from a disadvantage is 7.4 percentage points lower. The entrant pool therefore shifts away from participants at a competitive disadvantage without an increase in the share who compete with an advantage. Participants reporting a disadvantage are also more likely to have reported honestly.[1] Limited verification may therefore reduce the representation of honest reporters. The institutional pattern is similar for women and men.

A conditional comparison reveals a further pattern. Among participants holding a competitive advantage, entry is 12.2 percentage points higher after random assignment than after a private report. Selection into the privately reported easy task may contribute to this difference.

These findings contribute to research on competition, unequal playing fields, and private information. Studies of competition entry typically take the source of competitive position as given, while studies of unequal playing fields vary the position itself (Trautmann, 2023; Buser et al., 2025). We link these literatures by showing that how a competitive position is determined changes entrant composition. We also connect private reporting to a subsequent competition decision. The reporting stage creates a selection margin that is absent under random assignment without requiring us to classify individual participants as truthful or dishonest (Gneezy, 2005; Fischbacher and Föllmi-Heusi, 2013; Schwieren and Weichselbaumer, 2010; Faravelli et al., 2015; Buser and Sangi, 2025). The practical implication is that verification can shape access to competition, not only the credibility of the claims that determine advantage.

The remainder of the paper is organized as follows. Section 2 discusses the related literature. Section 3 describes the experimental design. Section 4 develops the hypotheses. Section 5 reports the main results. Section 6 examines what aggregate reporting can tell us about selection into competitive advantage. Section 7 concludes.

[1] If participants do not report hard after rolling easy, all hard reports are truthful (Houser et al., 2012).

2 Related Literature

We contribute to the literature on willingness to compete. The canonical tournament design relates entry to expected relative performance, confidence, risk attitudes, and preferences for competition, and documents substantial gender differences (Gneezy et al., 2003; Niederle and Vesterlund, 2007; Croson and Gneezy, 2009; Niederle and Vesterlund, 2011; van Veldhuizen, 2022). These differences vary across settings and can change when the structure or purpose of competition changes (Healy and Pate, 2011; Apicella et al., 2017; Cassar and Rigdon, 2021; Markowsky and Beblo, 2022). Much of this work explains entry through individual characteristics or changes to the structure and purpose of competition. We study a complementary institutional margin: how a participant's competitive position is determined. Our design also separates absolute task difficulty from relative competitive position. An easier task raises expected output under both payment schemes, whereas a competitive advantage changes expected performance relative to the opponent.

This institutional focus connects our paper to research on procedural fairness and uneven playing fields. Research on procedural fairness shows that people care about how opportunities are allocated, not only final outcomes; for a review, see Trautmann (2023). More directly, Buser et al. (2025) examine willingness to compete on uneven playing fields. We extend this work by examining how an uneven playing field arises. In our design, a competitive advantage is either assigned by computer or follows a private report that may favor the participant. We then examine whether these institutions produce different patterns of entry from each competitive position.

The private reporting institution further connects our study to research on dishonesty and competition. Because individual misreporting is usually unobservable, aggregate reports are used to detect departures from a truthful benchmark (Gneezy, 2005; Fischbacher and Föllmi-Heusi, 2013). Other studies examine how competition affects misconduct and willingness to enter environments in which competitors may behave strategically (Schwieren and Weichselbaumer, 2010; Faravelli et al., 2015; Buser and Sangi, 2025). Our design joins these questions by placing a private report before competition entry. The report determines competitive position, allowing us to study how an unverifiable claim shapes subsequent selection into competition while the tournament rule remains fixed.

3 Experimental Setup

3.1 Design

We conducted a controlled laboratory experiment in which participants decide whether to enter a tournament from equal or unequal competitive positions. The design varies both the participant's position and whether an unequal position arises through computer assignment or private reporting.

The experiment features four treatments: *Die Roll*, *Random*, *Fixed Easy*, and *Fixed Hard*. Die Roll and Random create unequal competitive positions. In Die Roll, participants privately report the task assignment implied by their die roll. In Random, the computer assigns task difficulty by chance. Fixed Easy and Fixed Hard provide equal task benchmarks in which all participants perform the same task. All four treatments use variants of the tournament task in Niederle and Vesterlund (2007).

Each treatment comprises three parts followed by a questionnaire. Parts One and Two are identical across treatments; only Part Three varies.

Part One

In Part One, participants have 30 seconds to solve as many addition problems with single digit numbers as possible (see Figure 1a). They earn 20 pence per correct answer.

Part Two

In Part Two, participants have another 30 seconds to solve arithmetic problems involving both addition and subtraction (see Figure 1b). On average, they solve about half as many problems correctly as in Part One, making Part Two substantially more difficult.

(a) Part One task example

(b) Part Two task example

Figure 1: Examples of tasks used in Part One and Part Two

Parts One and Two measure baseline ability and give participants direct experience of the difference in task difficulty before they choose their payment scheme in Part Three.

Part Three

In Part Three, we randomly pair participants and ask them to choose between piece rate and tournament compensation. Under the piece rate, participants earn 20 pence per correct answer. Under the tournament, they earn 60 pence per correct answer if they outperform their partner, 30 pence if they tie, and zero if they solve fewer problems correctly. Tournament outcomes depend only on the number of correct answers, regardless of which task each partner performs. Participants again have 30 seconds to complete the task.

To limit prosocial concerns toward the partner, we tell participants that their compensation choice affects only their own payoff. A participant may enter the tournament even if the partner chooses piece rate. The entrant's earnings then depend on relative performance, while the partner receives piece rate compensation.

The task assigned to each participant in Part Three varies across treatments:

- **Die Roll**: Each participant receives a fair six-sided die and rolls it twice but reports only the outcome of the first roll (Fischbacher and Föllmi-Heusi, 2013). The reported roll determines the task: outcomes of 1–3 correspond to the easier Type A task from Part One, while outcomes of 4–6 correspond to the harder Type B task from Part Two. Reporting is private and unverifiable, allowing participants to report an outcome that favors them. Participants report the roll, learn their task, and choose their compensation scheme on the same page (see Figure 2).

- **Random**: The computer assigns participants to either the Type A or Type B task. We calibrated the assignment probabilities using the reporting distribution observed in a pilot of Die Roll.[2] Participants learn their task before choosing between piece rate and tournament (see Figure 3), but they do not observe the assignment probabilities.

- **Fixed Easy**: All participants learn that everyone will perform the Type A task from Part One before choosing their compensation scheme.

- **Fixed Hard**: All participants learn that everyone will perform the Type B task from Part Two before choosing their compensation scheme.

[2] We used the pilot share reporting Type A to set the probability of each task in Random.

Throughout the paper, we use easy and hard to describe the task itself. We use *competitive advantage* and *competitive disadvantage* to describe a participant's relative position in Random and Die Roll, where matched partners may perform different tasks. Participants performing the easy task hold a competitive advantage, while those performing the hard task hold a competitive disadvantage. Fixed Easy and Fixed Hard are equal task benchmarks because both partners perform the same task. Random and Die Roll differ in how the task, and hence the competitive position, is determined: the computer assigns the task in Random, while the task follows a private report in Die Roll.

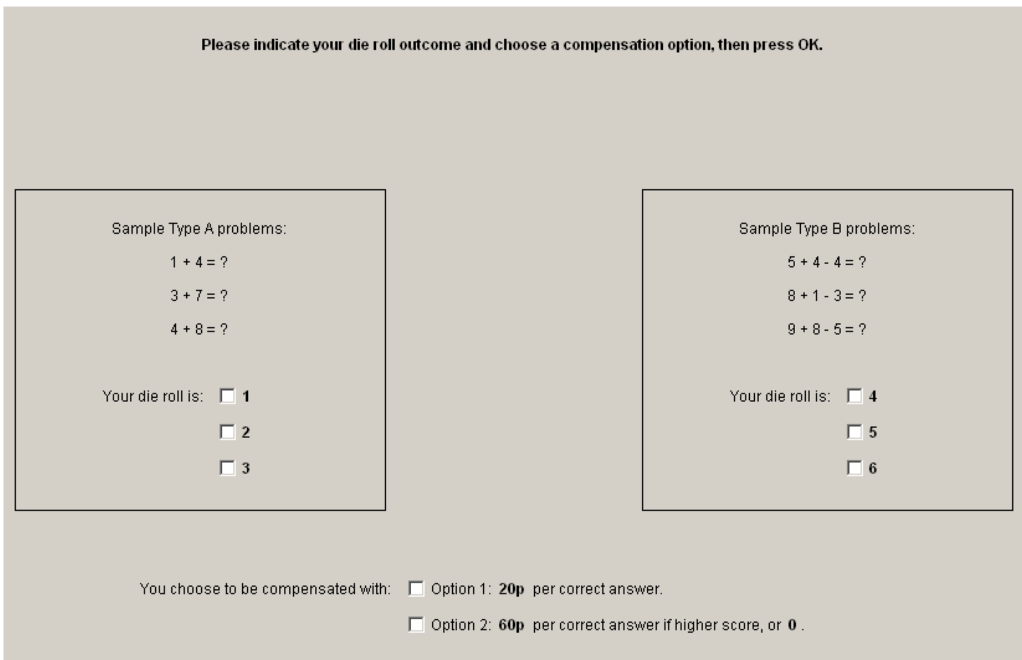

Figure 2: Die Roll Treatment: Decision Page

After Part Three, participants make incentivized guesses and complete a questionnaire. We randomly select one of the three incentivized parts for payment at the end of the experiment. Participants report beliefs about average performance on both tasks and about the matched partner's competition entry. In Random, they report partner entry conditional on assignment to each task. In Die Roll, they also guess which task the partner reported. We do not disclose the assignment probabilities in Random or the distribution of reports in Die Roll, so beliefs about the partner's task are formed within each institutional environment. Figure 4 summarizes the experiment.

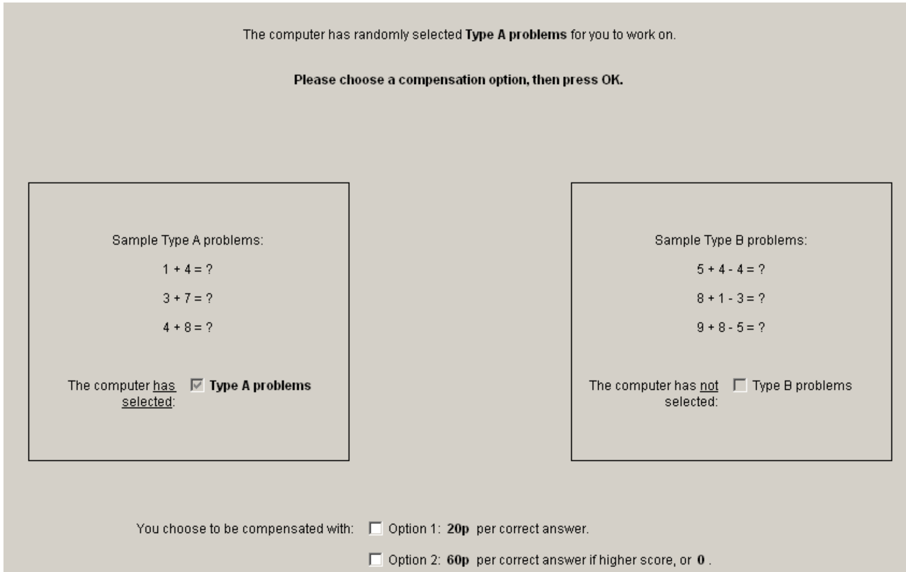

Figure 3: Random Treatment: Decision Page

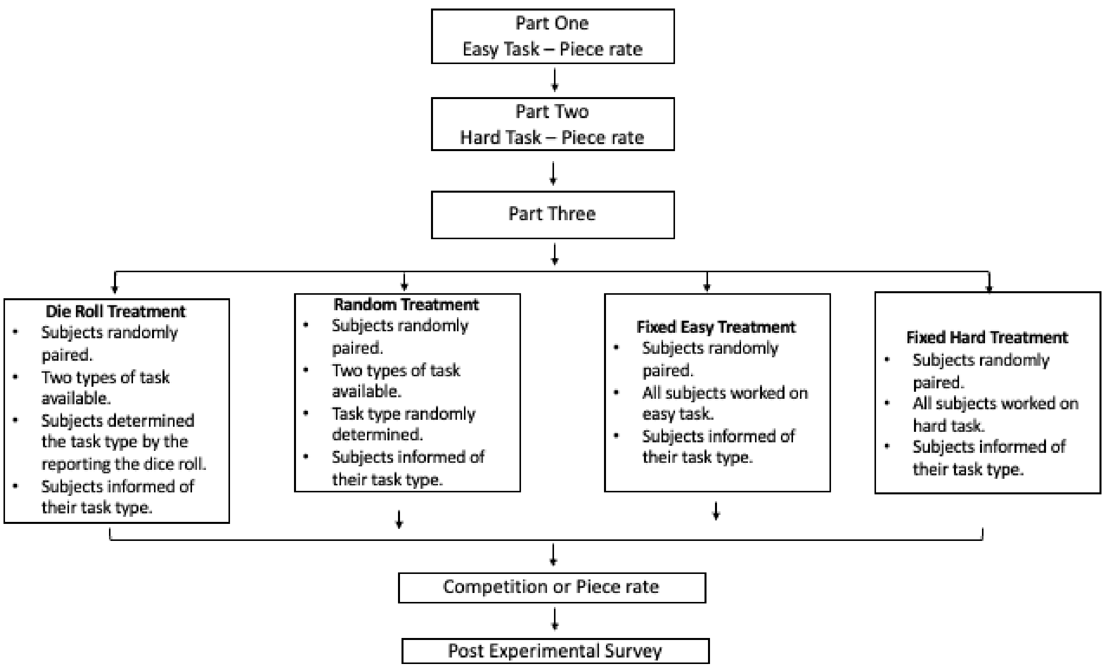

Figure 4: Timeline of the Experimental Treatments

3.2 Procedures

We conducted the experiment in the Finance and Economics Experimental Laboratory at Exeter (FEELE) between December 2017 and March 2018. We recruited 654 participants from the laboratory's subject database using ORSEE (Greiner,

2004) and ran 26 sessions with 14 to 30 participants each. Participants sat at individual stations with dividers, and the experimenter read the instructions aloud. Sessions lasted about 45 minutes. Mean earnings were approximately £10, including a £3 show up fee.

We programmed the experiment in z-Tree (Fischbacher, 2007). It comprised three incentivised parts and a demographic survey. Participants knew that we would randomly select one part for payment and received no feedback until the experiment ended. The complete experimental screens appear in the Online Appendix.

4 Conceptual Framework and Hypotheses

In this section, we present a simple framework that organizes the empirical analysis. The framework shows how entry can depend on absolute task difficulty, competitive position, and the institutional source of that position.

Consider a participant i who chooses between piece rate and tournament payment. Let $t_i \in \{E, H\}$ denote the participant's task, where E is the easy task and H is the hard task, and let $q_i(t_i)$ denote expected output on that task. Let s denote the competitive environment. The participant's beliefs $b_i(t_i, s)$ include beliefs about own and partner output, the partner's task or report, and the partner's entry decision. Let $p_i^W(t_i, s)$ and $p_i^T(t_i, s)$ denote the perceived probabilities of winning and tying the tournament. Under the piece rate, the participant receives 20 pence per correct answer. Under the tournament, the participant receives 60 pence per correct answer when outperforming the matched partner, 30 pence per correct answer in a tie, and zero when performing worse. The expected monetary gain from tournament rather than piece rate payment can therefore be represented as

$$M_i(t_i, b_i(t_i, s)) = q_i(t_i) \left[60p_i^W(t_i, s) + 30p_i^T(t_i, s) - 20\right]. \tag{1}$$

Equation (1) shows that the relationship between task difficulty and entry depends on expected relative performance. An easier task raises expected output under both payment schemes. Whether it increases the return to tournament entry therefore depends on the participant's perceived probabilities of winning and tying.

Competitive position changes this comparison through expected relative performance. In Fixed Easy, both competitors face the easy task; in Fixed Hard, both face the hard task. These conditions isolate task difficulty on a level playing field. In Random, a participant with the easy task may face a partner with the hard task and therefore holds a competitive advantage, while a participant with the hard task may face a partner with the easy task and therefore holds a competitive disadvantage. Comparing Random Easy with Fixed Easy and Random Hard with Fixed Hard holds the participant's task constant while varying whether it creates an unequal competitive position.

Hypothesis 1. Competitive position. *Relative to the corresponding equal task benchmark, a competitive advantage increases entry and a competitive disadvantage reduces entry.*

The institutional source of a competitive position can matter even when the observed task is held fixed. In Random, the computer assigns the task; in Die Roll, the task follows a private report. The institution can affect entry through beliefs about the partner's task, report, or entry decision. It may also change the nonmonetary value of exercising an advantage that was externally assigned or followed a private report.

We represent these channels in reduced form. Using this notation, participant i enters competition when

$$M_i(t_i, b_i(t_i, s)) + \omega_i(t_i, s) \geq 0, \tag{2}$$

where $\omega_i(t_i, s)$ is a nonmonetary component in reduced form. It may capture preferences for competition, risk attitudes not already reflected in the comparison of expected values, image concerns, moral or personal image costs, or responses to the institutional source of competitive position.

Random and Die Roll affect observed entry through two margins. First, computer assignment and private reporting can produce different shares of participants holding a competitive advantage or disadvantage. Second, competition entry can differ within each position. Together, these margins determine four joint outcomes: competitive advantage and compete, competitive advantage and piece rate, competitive disadvantage and compete, and competitive disadvantage and piece rate. The margins may offset, producing similar aggregate entry rates but different joint distributions of competitive position and competition entry.

Hypothesis 2. Institutional environment. *Random assignment and private reporting produce different joint distributions of competitive position and competition entry.*

The institutional source of an advantage can affect both $b_i(t_i, s)$ and $\omega_i(t_i, s)$ in

Equation (2). A participant who reports the easy task may expect others to do the same (Butler et al., 2015; Rauhut, 2013). This raises the perceived probability that the partner also performs the easy task and lowers the perceived probability of winning and the expected monetary gain from competition. The source can also change the nonmonetary value of entry. An advantage assigned by the computer does not make the participant responsible for creating the unequal position, whereas entering after a private report means benefiting from an advantage based on the participant's own report. If this carries a moral or personal image cost, entry becomes less attractive (Gneezy et al., 2018; Abeler et al., 2019). Both channels predict that participants will exercise an externally assigned advantage more readily than one that follows a private report.

Hypothesis 3. Exercising competitive advantage. *Among participants holding a competitive advantage, competition entry is higher when the advantage is externally assigned than when it follows a private report.*

5 Results

We begin by summarizing the sample, baseline performance, and the distribution of participants across the competitive environments. We then present the comparisons organized by the three hypotheses in Section 4.

5.1 Summary Statistics

Our sample includes 654 participants with an average age of 19.9 years; 54.0 percent are women. Performance confirms the difference in task difficulty. Participants solve an average of 17.6 problems in Part One and 10.0 in Part Two. The gap persists in Part Three: participants performing the easy task solve 21.3 problems on average, compared with 10.8 among those performing the hard task. Appendix Table A.1 reports performance and entry by treatment and the task performed in Part Three.

The treatments differ in whether matched competitors perform the same task and in how their tasks are determined. In Fixed Easy and Fixed Hard, both competitors perform the same task. In Random and Die Roll, matched competitors may perform different tasks. In the realized sample, 65.0 percent of participants in Random were assigned to the easy task, while 79.7 percent in Die Roll reported an outcome that led to the easy task. Appendix Table A.2 reports the full assignment and reporting shares.

The differences in task composition mean that the overall entry rate in Random and Die Roll combines participants in different competitive positions. We refer to the share choosing competition within each treatment as aggregate competition entry. This rate is 58.8 percent in Fixed Easy, 53.3 percent in Fixed Hard, 58.3 percent in Random, and 52.9 percent in Die Roll. Although these rates span only 5.9 percentage points, similar aggregate rates can mask offsetting changes across competitive positions. In Sections 5.2 and 5.3, we unpack these changes.

5.2 Causal Effects of Advantage and Disadvantage

Competitive advantage increases competition entry, while competitive disadvantage reduces it. The response is concentrated among men. Hypothesis 1 predicts these opposing responses. To test this hypothesis, we use only Random, Fixed Easy, and Fixed Hard. We compare Random Easy with Fixed Easy and Random Hard with Fixed Hard, holding the participant's task fixed while varying whether it creates an unequal competitive position.

Figure 5 presents the raw comparisons, first for the full sample and then separately for women and men. In the pooled sample, entry rises when the easy task creates a competitive advantage and falls when the hard task creates a competitive disadvantage. The gender panels show that these responses are driven mainly by men. Women's entry remains close to the corresponding equal task benchmarks, while men respond strongly to both competitive advantage and disadvantage.

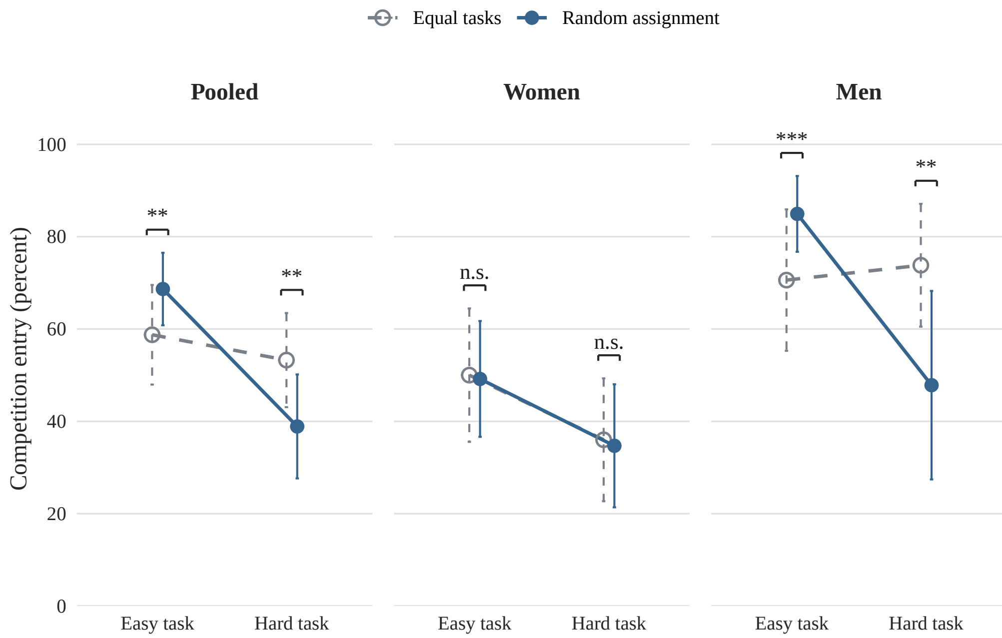

Figure 5: Competition Entry under Equal Tasks and Random Assignment, Pooled and by Gender

Notes: Points report raw competition entry rates for the full sample and separately for women and men. Lines connect the easy and hard tasks within each environment. Equal tasks refers to Fixed Easy and Fixed Hard. Random assignment refers to participants assigned the corresponding task in Random. Vertical bars denote 95 percent confidence intervals. Brackets report the estimated comparisons in Table 1. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; n.s. denotes $p > 0.10$.

The elicited beliefs support this interpretation. In Random, 85.0 percent of participants predicted that a partner assigned the easy task would enter competition, compared with 15.0 percent when the partner was assigned the hard task. Participants therefore anticipated that the easier task would make competition considerably more attractive, consistent with our interpretation of the easy task as a competitive advantage and the hard task as a competitive disadvantage (Appendix Table A.3).

To determine whether this pattern remains after accounting for participant characteristics, we estimate one pooled linear probability model:

$$\begin{aligned} C_i = \alpha + \delta E_i + \theta R_i + \mu M_i + \lambda(E_i \times R_i) \\ + \phi(E_i \times M_i) + \psi(R_i \times M_i) + \kappa(E_i \times R_i \times M_i) + X_i'\gamma + \varepsilon_i, \end{aligned} \tag{3}$$

where C_i indicates competition entry, E_i indicates the easy task, R_i indicates Random, and M_i indicates men. The vector X_i includes performance in Parts One and Two, risk preferences, stated competitiveness, age, and math experience. From this model, we estimate the effect of competitive advantage relative to Fixed Easy, the effect of competitive disadvantage relative to Fixed Hard, and the difference between these responses. We report each comparison for women and men and average them using their shares in the estimation sample for the pooled results.

Table 1 reports these contrasts. In the pooled sample, competitive advantage raises entry by 13.0 percentage points ($p = 0.037$), while competitive disadvantage lowers entry by 15.3 percentage points ($p = 0.032$). These similarly sized movements in opposite directions widen the easy versus hard entry gap by 28.3 percentage points ($p = 0.003$).

Table 1: Competitive Advantage, Disadvantage, and Competition Entry

	Pooled	Women	Men	Male minus female
Competitive advantage	13.0**	3.9	23.9***	20.0
	(6.2)	(9.1)	(8.1)	(12.3)
Competitive disadvantage	-15.3**	-8.5	-23.5**	-15.0
	(7.1)	(9.7)	(11.0)	(14.9)
Advantage minus disadvantage	28.3***	12.4	47.4***	35.0*
	(9.4)	(13.4)	(13.4)	(19.3)
Controls			Yes	
Observations			378	

Notes: All entries are contrasts from one pooled linear probability model with indicators for the easy task, Random, and male, including all interactions among these indicators. Competitive advantage is Random Easy minus Fixed Easy, and competitive disadvantage is Random Hard minus Fixed Hard. Advantage minus disadvantage reports how much Random widens the easy versus hard entry gap relative to Fixed. The pooled column averages the gender specific contrasts using their estimation sample shares. The model controls for performance in Parts One and Two, risk preferences, stated competitiveness, age, and math experience. All differences and robust standard errors are expressed in percentage points, with standard errors shown in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The pooled response is driven primarily by men. Competitive advantage raises men's entry by 23.9 percentage points, while competitive disadvantage lowers it by 23.5 percentage points. Together, these movements widen the entry gap among men by 47.4 percentage points. Among women, competitive advantage raises entry by 3.9 percentage points and competitive disadvantage lowers it by 8.5 percentage points; both changes are imprecisely estimated. The gender difference in the widening is 35.0 percentage points ($p = 0.070$).

Taken together, the opposing responses show that entry depends on whether a task creates a competitive advantage or disadvantage, not simply on its absolute difficulty.

Result 1 *The evidence supports Hypothesis 1: competitive advantage increases entry, while competitive disadvantage reduces it. This response is concentrated among men; women's entry changes little across the corresponding environments.*

5.3 Causal Effect of the Institutional Environment

We next turn to Hypothesis 2, which predicts that Random and Die Roll produce different joint distributions of competitive position and competition entry. We use only these two treatments. Because the institutional environment was randomly assigned, the comparison identifies the causal effect of private reporting relative to random assignment. The joint distribution captures both the competitive position a participant reaches and whether the participant competes from that position.

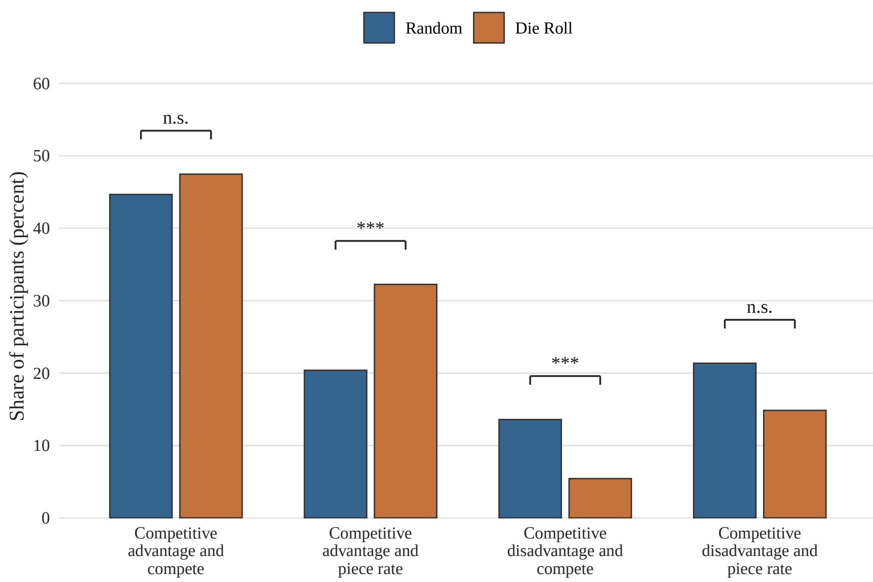

Figure 6: Joint Distribution of Competitive Position and Competition Entry
Notes: Competitive advantage denotes the easy task and competitive disadvantage denotes the hard task within Random and Die Roll. Bars report the raw percentage of participants in each joint outcome. The four outcomes are mutually exclusive and sum to 100 percent within each treatment. Brackets report the estimated differences in Table 2. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; n.s. denotes $p > 0.10$.

Figure 6 shows a clear reallocation across the four outcomes. On the advantage side, the share who compete changes little, from 44.7 percent in Random to 47.5 percent in Die Roll. The larger difference appears among participants who hold an advantage and choose piece rate, whose share rises from 20.4 to 32.2 percent. Under private reporting, more participants hold an advantage without a comparable increase in the population share competing with one.

On the disadvantage side, the share who compete falls from 13.6 percent in Random to 5.4 percent in Die Roll, while the share who choose piece rate falls from 21.4 to 14.9 percent. The institution therefore reduces the population share that enters from a disadvantage. Together, the changes in competitive position and competition entry capture the causal effect of the assigned institutional environment. Section 5.4 then examines entry within each position.

To test whether these differences remain after accounting for participant characteristics, we estimate a separate linear probability model for each joint outcome:

$$Y_{ij} = \alpha_j + \beta_j D_i + X_i' \gamma_j + \varepsilon_{ij}, \tag{4}$$

where Y_{ij} indicates whether participant i realizes joint outcome j, and D_i indicates assignment to Die Roll rather than Random. The vector X_i contains the participant controls defined in Section 5.2. Table 2 reports the estimated Die Roll minus Random difference in each outcome share. We use a multinomial logit model with the same controls to test whether the two institutions produce the same joint distribution.

Table 2: Institutional Differences in Competitive Position and Competition Entry

Joint outcome	Die Roll − Random (percentage points)
Competitive advantage and compete	1.2
	(4.4)
Competitive advantage and piece rate	12.2***
	(3.9)
Competitive disadvantage and compete	-7.4***
	(2.7)
Competitive disadvantage and piece rate	-6.0
	(3.6)
Controls	Yes
Observations	482
H_0: Random and Die Roll have the same joint distribution	$p = 0.002$

Notes: Competitive advantage denotes the easy task and competitive disadvantage denotes the hard task within Random and Die Roll. Entries report estimated Die Roll minus Random differences in percentage points from separate linear probability models for the listed joint outcomes. All models control for baseline task performance, risk preferences, stated competitiveness, age, and math experience. Robust standard errors are in parentheses. The joint p-value is from a likelihood ratio test of the Die Roll indicator in a multinomial logit model of the four outcomes with the same controls. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Under private reporting, the share with a competitive advantage who chooses piece rate is 12.2 percentage points higher than under random assignment ($p = 0.002$), while the share with a competitive advantage who competes differs by only 1.2 percentage points ($p = 0.794$). The share who both face a competitive disadvantage and compete is 7.4 percentage points lower under private reporting ($p = 0.007$). A joint test rejects equality of all four outcomes across the two institutional environments ($p = 0.002$).

The raw entry rate is 58.3 percent in Random and 52.9 percent in Die Roll, a difference of 5.4 percentage points. This aggregate difference misses the main institutional change. The population share entering with an advantage remains stable, more participants hold an advantage but choose piece rate, and fewer participants enter from a disadvantage. Private reporting therefore changes which competitive positions are represented among entrants, not simply the total number who enter. The same pattern appears for women and men, with no clear difference between genders (Appendix Table A.4).

Result 2 *The evidence supports Hypothesis 2: private reporting changes the joint distribution of competitive position and competition entry. Relative to random assignment, it leaves the share who compete with an advantage unchanged, increases the share who hold an advantage and choose piece rate, and reduces the share who compete from a disadvantage.*

5.4 Exercising Competitive Advantage

In the previous section, we show that private reporting changes the joint distribution of competitive position and competition entry. To understand this change, we separate differences in task composition from differences in entry within each position. Hypothesis 3 focuses on the second margin: participants are more likely to exercise a competitive advantage when it is externally assigned than when it follows a private report. Figure 7 presents the raw comparisons between Random and Die Roll. Among participants holding a competitive advantage, 68.7 percent enter after random assignment, compared with 59.5 percent after a private report. Entry also differs in the same direction among participants holding a competitive disadvantage, although this comparison uses a smaller group.

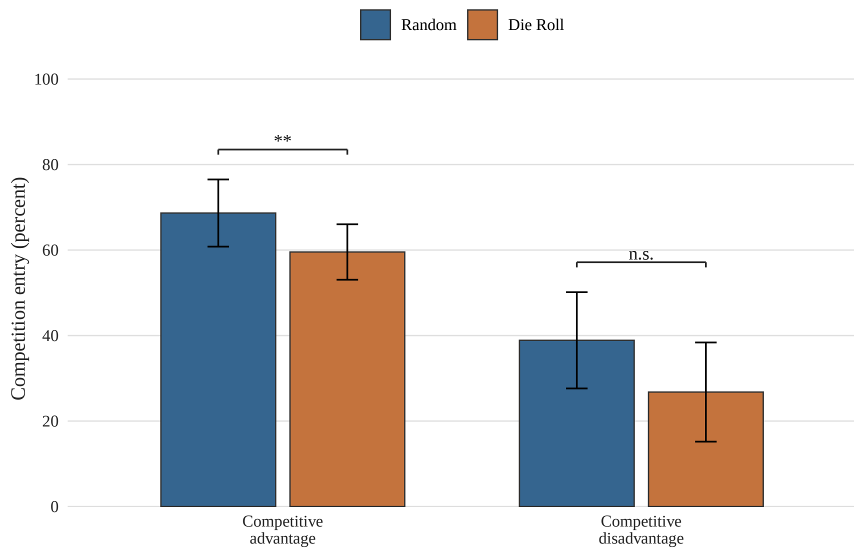

Figure 7: Competition Entry by Competitive Position and Institutional Environment

Notes: Bars report raw competition entry rates among participants holding the indicated competitive position in Random and Die Roll. Competitive advantage denotes the easy task and competitive disadvantage denotes the hard task. The task in Die Roll follows the participant's private report. Vertical bars denote 95 percent confidence intervals. Brackets report the estimated differences in Table 3. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; n.s. denotes $p > 0.10$.

For each task $t \in \{\text{easy}, \text{hard}\}$, we estimate

$$C_i = \alpha_t + \beta_t R_i + X_i'\gamma_t + \varepsilon_i, \tag{5}$$

where R_i indicates assignment to Random and X_i is defined as in Equation (3). Table 3 reports the estimated Random minus Die Roll difference within each competitive position. Appendix Table A.5 reports the comparisons separately for women and men.

Among participants holding a competitive advantage, entry is 12.2 percentage points higher after random assignment than after a private report ($p = 0.012$). The difference is larger among men, at 14.1 percentage points, than among women, at 6.4 percentage points, although the gender difference is not statistically clear ($p = 0.419$; Appendix Table A.5). The corresponding pooled difference among participants holding a competitive disadvantage is 3.9 percentage points ($p = 0.641$). The advantage comparison is therefore consistent with Hypothesis 3: participants are more likely to exercise a competitive advantage when it is externally assigned. Because the competitive advantage in Die Roll follows the participant's report, this comparison is descriptive; Section 6 examines the role of selection into the reported easy group.

Table 3: Institutional Source and Competition Entry by Competitive Position

	(1) Competitive advantage	(2) Competitive disadvantage
Random assignment − private reporting	12.2**	3.9
	(4.8)	(8.2)
Controls	Yes	Yes
Observations	354	128

Notes: The dependent variable is an indicator for competition entry. Each column reports the estimated Random minus Die Roll difference in percentage points from a linear probability model for participants holding the indicated competitive position. Competitive advantage denotes the easy task and competitive disadvantage denotes the hard task within Random and Die Roll. These comparisons hold the task fixed; the task in Die Roll follows the participant's private report. Controls include baseline task performance, risk preferences, stated competitiveness, age, and math experience. Robust standard errors are shown in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Result 3 *The conditional evidence is consistent with Hypothesis 3. Among participants holding a competitive advantage, competition entry is higher when the advantage is externally assigned than when it follows a private report.*

6 Selection into Competitive Advantage

Result 3 conditions on holding a competitive advantage, but the private report determines who enters that group in Die Roll. The lower entry rate after a private report may therefore reflect selection into competitive advantage, a response to the institution, or both. Aggregate reports reveal the implied composition of the easy report group under the reporting model of Houser et al. (2012), but connecting reporting types to competition entry requires further restrictions.

Under the reporting model, the 79.7 percent easy report rate implies that 62.7 percent of easy reports are truthful and 37.3 percent are false. Without restricting entry by reporting type, entry among truthful easy reporters is bounded between 35.5 and 94.9 percent. This interval contains the 68.7 percent entry rate with advantage under random assignment, so aggregate reporting cannot rule out selection as an explanation for Result 3. Appendix Table A.6 presents the calculation and gender specific bounds. Our institutional conclusion therefore rests on Result 2: private reporting changes the joint distribution of competitive position and competition entry.

7 Discussion and Conclusion

Competitive advantages do not all arise in the same way. In labor markets, promotion contests, procurement, and education, some reflect ability, training, or recognized qualifications. Others are created or amplified through access to networks, favorable assignments, strategic presentation of credentials, or claims that are difficult to verify. Much of the competition entry literature treats advantage as preexisting, arising for example from higher ability or prior performance, rather than as something institutions can produce (Niederle and Vesterlund, 2007, 2011; van Veldhuizen, 2022). Yet how an advantage is created can affect both who obtains it and who subsequently chooses to compete.

Research on uneven playing fields shows that externally created advantage and disadvantage affect willingness to compete (Trautmann, 2023; Buser et al., 2025). Our experiment moves closer to settings in which the institution helps produce the advantage itself. We compare an advantage or disadvantage assigned by the computer with one that follows a private report, while holding the tournament rule fixed. This allows us to study how the institution shapes both competitive advantage and the composition of entrants.

Our results first establish why competitive advantage and disadvantage matter for entry. Relative to equal task benchmarks, an advantage assigned by the computer raises entry by 13.0 percentage points, while a disadvantage reduces it by 15.3 percentage points. Participants respond to what the task implies for performance relative to a competitor, not simply to whether the task is easy or hard. These opposing responses are driven mainly by men.

The comparison between computer assignment and private reporting provides the central institutional result. Private reporting raises the share who hold an advantage but choose piece rate by 12.2 percentage points, leaves the share who compete with an advantage unchanged, and reduces the share who compete from a disadvantage by 7.4 percentage points. Total entry is only 5.4 percentage points lower. The principal change is therefore in the composition of competition: under private reporting, fewer participants enter from a disadvantage, while the larger share holding an advantage does not produce more advantaged entrants.

This composition reveals how limited verification can shape access to competition. Participants who report the hard outcome retain their disadvantage rather than claim an advantage. Their lower representation among entrants raises the possibility that institutions accepting favorable claims may screen out people who are less willing to make such claims. The welfare implications depend on what the reported disadvantage represents. A pool concentrated among participants with an advantage may improve sorting when disadvantage accurately reflects poor prospects in the contest. It may instead restrict access when disadvantage reflects a decision not to claim an advantage that others cannot readily verify. In either case, evaluating the institution requires examining who enters, whether they enter with an advantage or disadvantage, and how that advantage or disadvantage was obtained.

The gender findings reinforce the importance of studying the competitive environment rather than treating willingness to compete as a fixed individual characteristic. Men respond strongly when computer assignment creates an advantage or disadvantage, while women's entry remains close to the corresponding equal task benchmarks. The gender gap therefore depends on the environment in which entry occurs. This finding complements evidence that gender differences vary with the form and purpose of competition (Healy and Pate, 2011; Apicella et al., 2017; Cassar and Rigdon, 2021; Markowsky and Beblo, 2022) and with the compensation structures used to recruit applicants (Flory et al., 2015; Samek, 2019). At the same time, private reporting changes the joint distribution similarly for women and men. The institutional effect is broader than the gender difference in responses to advantage and disadvantage. Closing gender gaps in competitive participation is therefore not only a matter of changing women's preferences, confidence, or willingness to enter. It also requires examining the institutions in which competition takes place and the processes through which those institutions create advantage.

Our design preserves a central feature of many real competitions: claims that cannot be verified shape who obtains an advantage and who subsequently enters. This linkage is substantively important, but it limits what the experiment can separate. Among participants holding an advantage, entry is 12.2 percentage points higher after computer assignment than after a private report. Because reports determine advantage in Die Roll, this difference may reflect both selection and a response to private reporting, and the reporting bounds cannot rule out selection. Assigning false reports would separate these channels only by removing the reporting choice the experiment seeks to capture. We therefore treat the 12.2 percentage point difference as suggestive. The Random versus Die Roll comparison instead identifies the effect of the institution as a whole, including the selection it generates. We do not identify individual false reports or isolate the relative roles of selection, beliefs, and moral or personal image costs. Future work can distinguish these mechanisms by varying monitoring or disclosure more directly.

Our findings highlight an often overlooked feature of competitive design: how institutions create and verify advantage. More verification need not always improve welfare: it can be costly, and private claims may contain useful information. Yet verification affects more than the accuracy of those claims. It can change who remains disadvantaged and who reaches competition. Institutions can respond through clearer eligibility criteria, standardized documentation, selective audits, or independent verification when the expected benefit justifies the cost. They should also assess these procedures by the composition of applicants and entrants, not only by total participation. Policies that change tournament rules, provide training, or encourage entry address willingness to compete within an existing environment. Verification and disclosure shape the environment in which that decision is made.

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Appendix

A Additional Tables

Table A.1: Performance and Competition Entry by Treatment and Task

Treatment and task	N	Entry	Part One	Part Two	Part Three
Fixed Easy	80	58.8	18.5	9.7	22.4
Random: easy assigned	134	68.7	16.7	9.9	20.2
Die Roll: easy reported	220	59.5	18.4	10.0	21.7
Fixed Hard	92	53.3	17.6	10.1	11.5
Random: hard assigned	72	38.9	15.7	9.8	10.1
Die Roll: hard reported	56	26.8	17.8	10.3	10.4

Notes: Statistics are reported by treatment and the task performed in Part Three. Entry is the percentage choosing tournament payment. Parts One and Two are the easy and hard tasks completed before this decision; the three performance columns report the mean number of correct answers.

Table A.2: Task Assignment and Reporting Shares

Group	Fixed Easy	Fixed Hard	Random	Die Roll
Panel A: Share with the easy task (percent)				
Overall	100.0	0.0	65.0	79.7
Women	100.0	0.0	55.5	75.5
Men	100.0	0.0	76.0	84.5
Panel B: Observations				
Overall	80	92	206	276
Women	46	50	110	147
Men	34	42	96	129

Notes: Panel A reports the percentage of participants assigned to the easy task in Fixed Easy, Fixed Hard, and Random, and the percentage reporting the easy task in Die Roll. Panel B reports observation counts. Fixed Easy and Fixed Hard are compliance checks because task type is fixed by design. In Random, the probabilities of receiving the easy or hard task were calibrated using the reporting distribution from the Die Roll pilot.

Table A.3: Beliefs about Partner Competition Entry in Random

	Predicted entry (percent)
Partner assigned the easy task	85.0
	(2.5)
Partner assigned the hard task	15.0
	(2.5)
Easy minus hard	69.9***
	(4.1)
Observations	206

Notes: Participants in Random predicted whether their matched partner would choose competition under each possible task assignment. The first two rows report the shares predicting entry. Easy minus hard is the within participant difference, with standard errors in parentheses. Beliefs were elicited after participants made their own entry decisions and are used to assess whether they perceived the competitive implications of task assignment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.4: Institutional Differences in Competitive Position and Competition Entry by Gender

Joint outcome	Women	Men	Men − women
Competitive advantage and compete	7.2	-5.0	-12.3
	(5.7)	(6.3)	(8.4)
Competitive advantage and piece rate	11.8**	12.3**	0.5
	(5.8)	(4.8)	(7.5)
Competitive disadvantage and compete	-8.0**	-6.7*	1.3
	(4.0)	(3.7)	(5.5)
Competitive disadvantage and piece rate	-11.0**	-0.5	10.4
	(5.5)	(4.4)	(6.9)
Controls		Yes	
Observations	257	225	482

Notes: Entries for women and men report estimated Die Roll minus Random differences in percentage points. The final column reports the difference between the estimates for men and women. Each row is based on a linear probability model for the listed joint outcome with indicators for Die Roll and male and their interaction. All models include the controls used in Table 2. Robust standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.5: Competition Entry by Institutional Source, Competitive Position, and Gender

	Women	Men	Pooled
Panel A. Competitive advantage			
Advantage under random assignment versus easy benchmark	4.6	25.1***	14.8**
	(9.2)	(8.4)	(6.2)
Advantage under private reporting versus easy benchmark	0.3	11.1	5.4
	(7.9)	(7.8)	(5.6)
Random assignment versus private reporting, advantage	6.4	14.1**	10.4**
	(7.6)	(5.7)	(4.7)
Panel B. Competitive disadvantage			
Disadvantage under random assignment versus hard benchmark	-9.4	-25.1**	-15.6**
	(9.8)	(11.4)	(7.1)
Disadvantage under private reporting versus hard benchmark	-6.8	-37.5***	-19.6***
	(9.9)	(11.2)	(7.4)
Random assignment versus private reporting, disadvantage	-0.7	9.9	2.9
	(10.2)	(13.7)	(8.2)

Notes: The dependent variable is competition entry. Entries are estimated comparisons expressed in percentage points; standard errors are reported in parentheses. Specifications are linear probability models with robust standard errors and controls for baseline task performance, risk preferences, stated competitiveness, age, and math experience. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.6: Reporting Mixture and Competition Entry Under Private Reporting

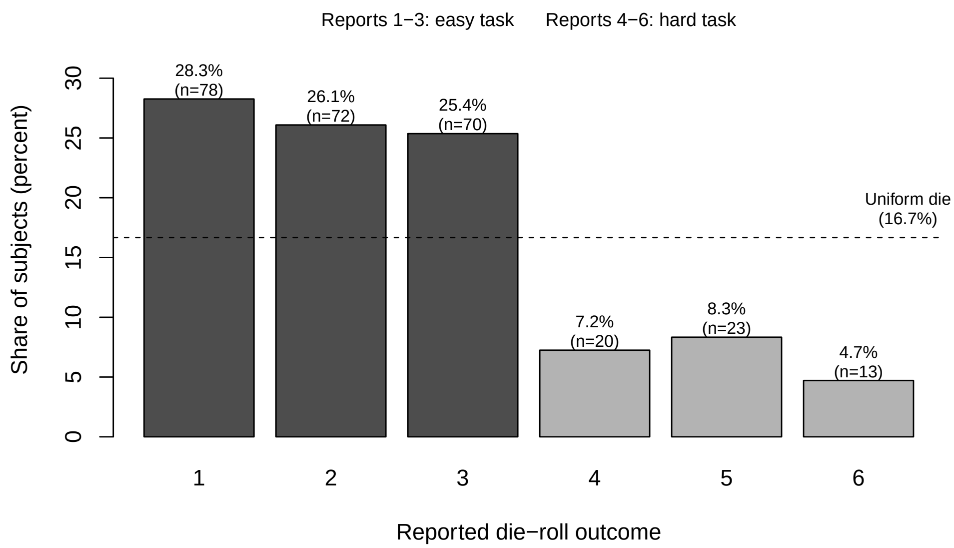
	Pooled	Women	Men
Panel A. Reporting composition			
Share reporting the easy task (q)	79.7	75.5	84.5
Implied always easy reporting type share ($2q-1$)	59.4	51.0	69.0
Implied share of false easy reports ($q-0.5$)	29.7	25.5	34.5
Truthful share among easy reports ($0.5/q$)	62.7	66.2	59.2
Panel B. Competition entry			
Entry among easy reporters (p_E)	59.5	45.9	73.4
Unrestricted bounds on truthful easy entry (h)	[35.5, 94.9]	[18.4, 69.4]	[55.0, 100.0]
Entry with advantage under random assignment	68.7	49.2	84.9
Break even entry among false reporters (ℓ)	44.2	39.6	56.7
Observations, easy reporters	220	111	109
Observations, random advantage	134	61	73

Notes: All entries except observations are percentages. Following Houser et al. (2012), the reporting mixture assumes that some participants always report an easy outcome and all other participants report truthfully. The implied share of false easy reports equals the observed easy report share minus the 50 percent expected from a fair die. Under the same assumption, the truthful share among easy reports is $s = 0.5/q$. Observed entry among easy reporters satisfies $p_E = s\,h + (1-s)\,\ell$, where h is entry among truthful easy reporters and ℓ is entry among false easy reporters. The unrestricted bounds impose only $\ell \in [0, 1]$. Entry with advantage under random assignment refers to the Random condition, easy task. Break even entry is the value of ℓ that makes h equal to entry with advantage under random assignment.

Calculation: Allowing ℓ to range from zero to one gives the lower bound $h_L = \max\{0, [p_E - (1-s)]/s\}$ and the upper bound $h_U = \min\{1, p_E/s\}$. In the pooled sample, $q = 0.797$, $s = 0.627$, and $p_E = 0.595$, which produce $[h_L, h_U] = [0.355, 0.949]$ using the unrounded values. The break even entry rate is $\ell^* = (p_E - sh_R)/(1-s)$, where $h_R = 0.687$ is entry with advantage under random assignment; this calculation gives $\ell^* = 0.442$.

B Additional Figures

Figure B.1: Distribution of Reported Die Roll Outcomes in the Die Roll Condition



Notes: Bars report the share of Die Roll participants reporting each die outcome ($N =$ 276). Outcomes one through three correspond to the easy (Type A) task; outcomes four through six correspond to the hard (Type B) task. The dashed line marks the 16.7 percent share implied by truthful reporting of a fair die. A chi-square test rejects the uniform distribution, $\chi^2(5) = 99.3$, $p < 0.001$.